GateMate™ FPGA User Guide

Interface Guide for Peripheral Devices with 3.3 V Signaling



UG1003

User Guide August 2025

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About this Document

This User Guide covers the interfacing of the Cologne Chip GateMate™ FPGA Series to peripheral devices with 3.3 V signaling and is part of the GateMate™ documentation collection.

For more information please refer to the following documents:

- Technology Brief of GateMate™ FPGA
- **DS1001** – GateMate™ FPGA CCGM1A1 Datasheet
- **UG1002** – GateMate™ FPGA Toolchain Installation User Guide

Cologne Chip provides a comprehensive technical support. Please visit our website for more information or contact our support team.

Revision History

This User Guide is constantly updated. The latest version of this document can be found following the link below:

> **UG1003** – GateMate™ FPGA Interface Guide for Peripheral Devices with 3.3 V Signaling

Date	Remarks
August 2025	Connection example using `LSF0108` level shifter added in Section 3.2.
December 2023	Initial release.

1 Introduction

This document describes how GateMate™ FPGAs can be connected to peripheral devices with 3.3 V signaling.

GateMate's GPIO supply voltage VDDIO is not allowed to exceed 2.7 V. Input voltages of 3.3 V or higher are not supported from the GateMate FPGA on chip level.

The single-ended general purpose input / output (GPIO) support the low voltage CMOS (LVCMOS) operating mode compliant to the standard JESD8-7(A), which means that the GPIO voltage can be 1.8 V ±0.15 V (normal range) or 1.2 V . . 1.95 V (wide range)

Any voltage above VDDIO supply voltage should be avoided on the GateMate pins.

2 Unidirectional Interfacing

2.1 GateMate Output to Peripheral Input

In cases where 2.5 V is used as GateMate VDDIO voltage and the peripheral receiver is an LVCMOS input with 3.3 V signaling, no interface circuitry is needed as shown in Figure 1.

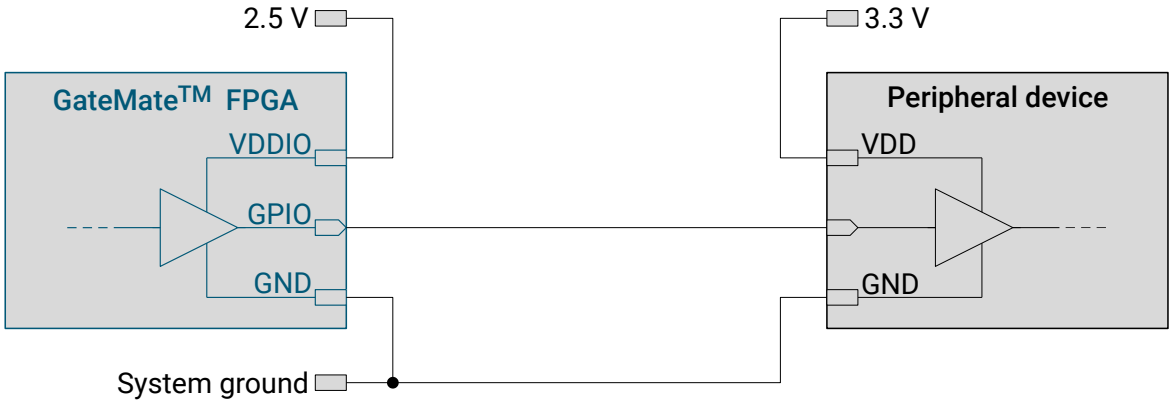


Figure 1: *GateMate output interfacing*

2.2 GateMate Input from Peripheral Output

If the peripheral driver connected to a GateMate input pin is driving 3.3 V and GateMate VDDIO voltage is 2.5 V, an interface circuitry is needed.

2.2.1 Simple Interface for Low Speed Signals

For low speed signals a simple series resistor is sufficient as shown in Figure 2. The input overvoltage security circuitry of the GateMate input pin will limit the input voltage.

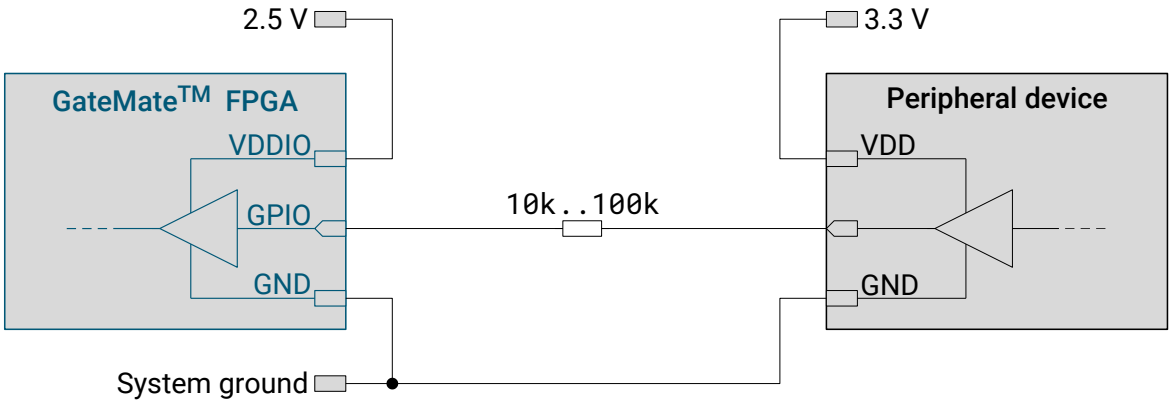


Figure 2: *Low speed GateMate input interfacing*

2.2.2 Interfacing for High Speed Signals

For high speed signal transmission an active interface circuit is recommended. The solution shown in Figure 3 has a delay of less than 10 ns and is therefore also suitable for high speed communication. The 74LV245 or 74LVC245 are 5 V tolerant devices that can be used with 2.5 V supply voltage and are able to have an input voltage of up to 5 V.

> **Please note!**
>
> Please make sure that you use an overvoltage-tolerant device, as the characteristics of the devices from various chip manufacturers may differ.

Hint: Figure 3 may also be an alternative to Figure 2 for low-speed signal buses in order to reduce the assembly costs of the PCB.

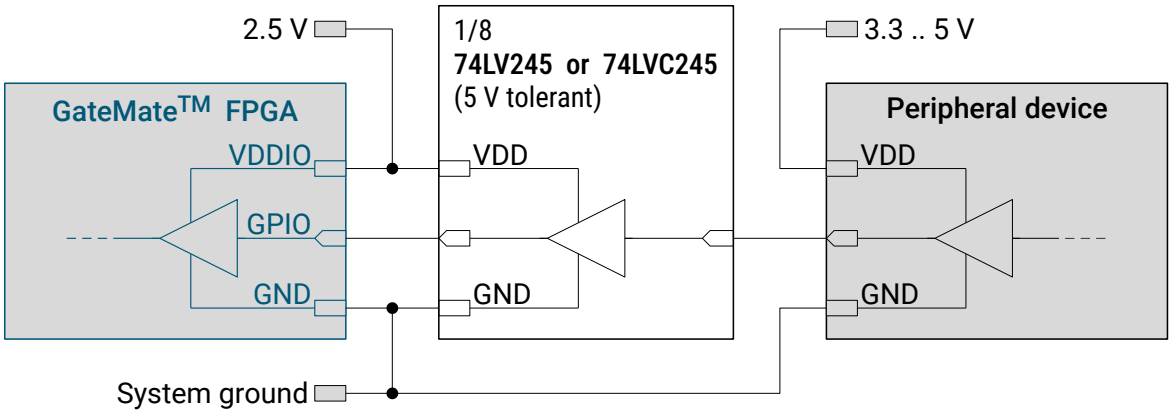


Figure 3: *High speed GateMate input interfacing*

3 Bidirectional Interfacing

In applications in which a GateMate FPGA is connected bidirectionally to peripheral devices, two cases must be distinguished:

1. There is a direction signal available to determine the direction. If it is a bus, all signals always have the same direction.
2. No direction signal is available or it is a bus where the individual signals can have different directions.

3.1 Bidirectional interfacing with direction signal

If a direction signal is available, a circuit with an active bidirectional buffer is recommended. The solution shown in Figure 4 has a delay of less than 10 ns and is therefore also suitable for high speed communication. The 74LV245 or 74LVC245 are 5 V tolerant devices that can be used with 2.5 V supply voltage and are able to have an input voltage of up to 5 V.

> **Please note!**
>
> Please make sure that you use an overvoltage-tolerant device, as the characteristics of the devices from various chip manufacturers may differ.

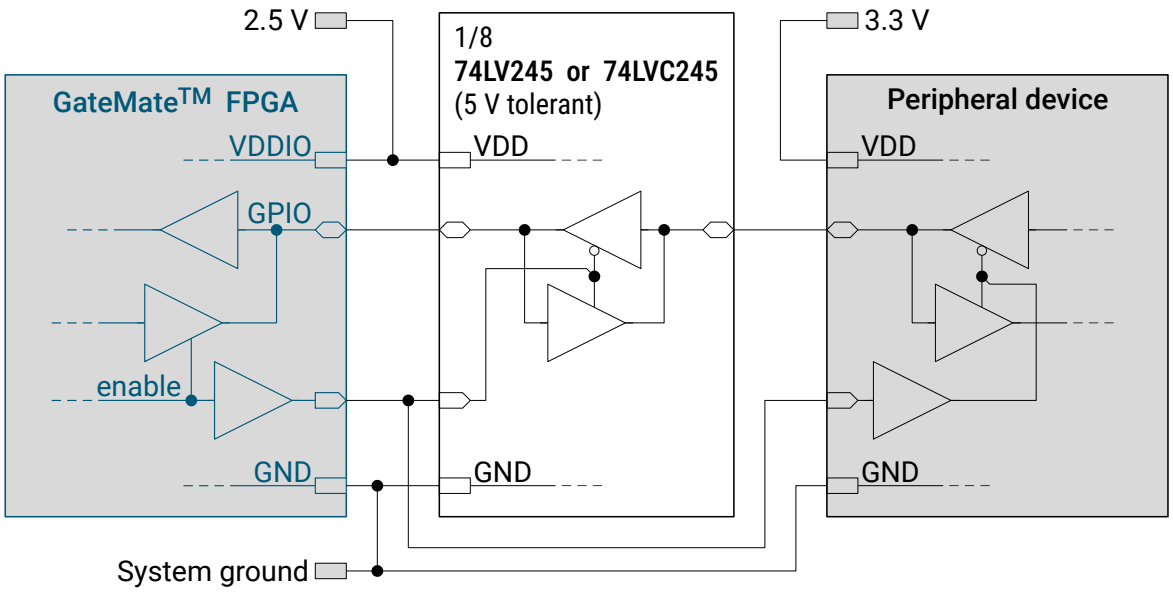


Figure 4: *Bidirectional high speed GateMate interfacing with direction signal*

3.2 Bidirectional interfacing without direction signal

In cases where no direction signal is available, there are devices which can limit the input voltage at both sides. The 8-bit FET bus switch device 74CB3T3245 has a low R_{on} resistance between the two terminals of the FET. This has a delay of less than 6 ns and is therefore also suitable for high speed communications. The 74CB3T3245 device must be used with a supply voltage of 2.5 V to limit the GateMate input voltage to 2.5 V as shown in Figure 5.

This proposed solution has good technical properties, but has higher component costs compared to the bus driver described in the previous section.

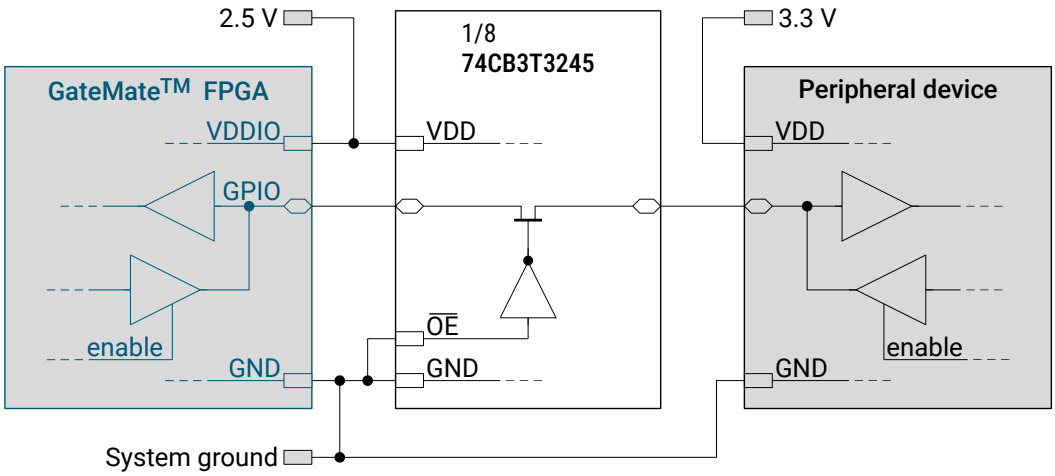


Figure 5: *Bidirectional high speed GateMate interfacing without direction signal using 74CB3T3245*

Alternatively, the LSF0108 device can also be used, for example. Figure 6 shows the principle of the connection with the GateMate FPGA.

LSF0108 is an auto-bidirectional 8-channel level shifter. It can be used for open-drain and push-pull applications. Please refer to the LSF0108 data sheet for the various connection types.

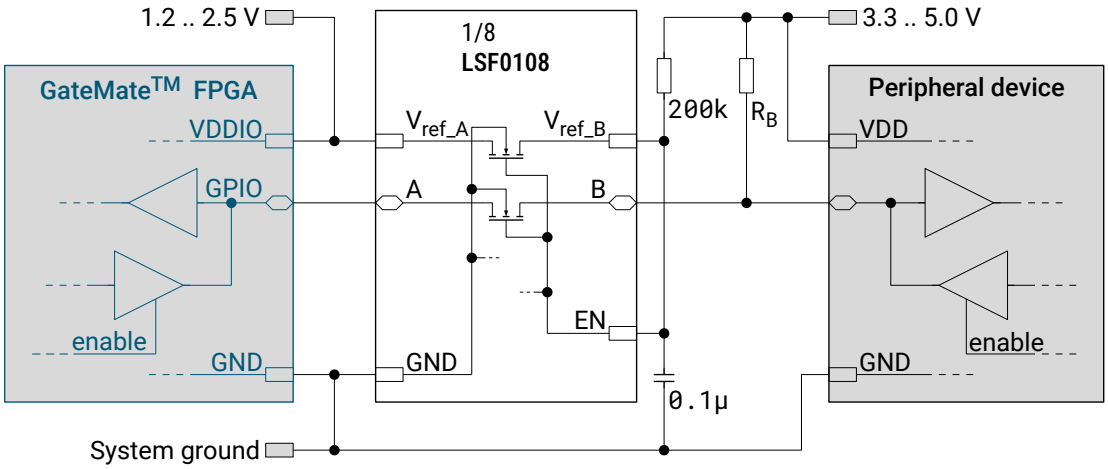


Figure 6: *Bidirectional high speed GateMate interfacing example using LSF0108*

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